

Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. (a) An insulated container of negligible heat capacity contains 1.5 kg of tea at a temperature of 60 °C.
- (i) What mass of ice at 0 °C should be added to the tea so that the final temperature of the mixture is lowered to 10 °C? Assume that the specific heat capacity of tea is the same as that of water. (3 marks)
- Given: specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$
specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

- (ii) If the heat capacity of the container is **not** negligible, explain whether more ice, less ice or the same amount of ice is needed to obtain the final temperature of 10 °C. (2 marks)

Answers written in the margins will not be marked.

Please stick the barcode label here.

- (b) Some ice cream at $-10\text{ }^{\circ}\text{C}$ is put into a 'thermal bag', of which the inner layer is made of polyethylene foam coated with aluminium foil. The bag is also equipped with a zipper at the top.



The thermal bag is then brought outdoors on a hot sunny day.

- (i) Referring to the heat transfer processes, explain ONE feature of this bag that helps keep the ice cream at a low temperature. (1 mark)

.....

.....

.....

- (ii) Suggest ONE modification to this bag that would enhance its ability to keep things stored inside at a low temperature. (1 mark)

.....

.....

.....

Answers written in the margins will not be marked.

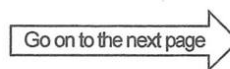
Answers written in the margins will not be marked.

Book 1 section 3.3: DSE long questions

2019 Q1 (page 3)

2019-DSE-PHY 1B-3

3



Provided by dse.life

Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. A 150 W immersion heater is used to keep the water in a large beaker boiling under standard atmospheric pressure. In 5 minutes, 16 g of water boils away. Neglect any heat loss to surroundings.

(a) Find the specific latent heat of vaporization of water, l . (2 marks)

.....

.....

.....

.....

.....

.....

A student puts a small metal sphere in the boiling water. After a few minutes, the sphere is quickly transferred to a polystyrene cup containing 100 g of water at a temperature of 20 °C. The cup of water is stirred gently and its highest temperature attained is 22 °C.

Given: specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹

(b) Estimate the heat capacity C of the metal sphere. (2 marks)

.....

.....

.....

.....

.....

.....

(c) In fact the sphere has carried with it some boiling water to the cup of water. Referring to this fact, explain whether the true value of C is higher or lower than the value calculated in (b). (2 marks)

.....

.....

.....

.....

.....

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Please stick the barcode label here.

- (d) In order to reduce the error contributed by the polystyrene cup, another student suggests repeating the measurements using a copper cup of similar shape and size. Explain whether the suggestion is justified. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81 \text{ m s}^{-2}$.

1. A mountaineer arrives at a location X , which is 3000 m above sea level. At X , the melting point of ice is 0°C and the boiling point of water is 90°C . The mountaineer needs to heat 0.50 kg of melting ice to its boiling point using a heater.

- (a) Estimate the energy required.

(2 marks)

Given: specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$
specific heat capacity of water = $4200 \text{ J kg}^{-1}^\circ\text{C}^{-1}$

Answers written in the margins will not be marked.

- (b) The power output of the heater is 2 kW. Estimate the time required.

(2 marks)

Answers written in the margins will not be marked.

- (c) Explain why the temperature of water remains at the boiling point after the water starts to boil, even when the heater is still on.

(1 mark)

Answers written in the margins will not be marked.

Book 1 section 3.3: DSE long questions

2026 Q1 (page 2)

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81 \text{ m s}^{-2}$.

1. A mountaineer arrives at a location X , which is 3000 m above sea level. At X , the melting point of ice is 0°C and the boiling point of water is 90°C . The mountaineer needs to heat 0.50 kg of melting ice to its boiling point using a heater.

- (a) Estimate the energy required.

(2 marks)

Given: specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$
specific heat capacity of water = $4200 \text{ J kg}^{-1}^\circ\text{C}^{-1}$

Answers written in the margins will not be marked.

- (b) The power output of the heater is 2 kW. Estimate the time required.

(2 marks)

Answers written in the margins will not be marked.

- (c) Explain why the temperature of water remains at the boiling point after the water starts to boil, even when the heater is still on.

(1 mark)

Answers written in the margins will not be marked.

Please stick the barcode label here.

*2. A sealed syringe contains neon gas of volume 5.0 cm^3 and pressure $1.0 \times 10^5 \text{ Pa}$. The gas is then compressed slowly to 2.0 cm^3 while the temperature remains at 27°C . Take the neon gas as an ideal gas.

(a) Find the pressure of the gas after being compressed. (2 marks)

(b) (i) Find the average kinetic energy of the gas molecules. (2 marks)

(ii) The mass of one mole of neon gas is 0.020 kg . Find the root-mean-square speed of the gas molecules. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Book 1 section 3.3: DSE long questions

2026 Q2 (page 3)

ANSWERS WRITTEN IN THE MARGINS WILL NOT BE MARKED.

2026-DSE-PHY 1B-3

3

